Hidden in the Leftovers

Mapping the Task-Specific Blind Spots of Sparse Autoencoders
in English and Indian Languages

Course project proposal · October 2026

In short. Sparse autoencoders (SAEs) are a leading tool for reading what a language model computes, and they pass the standard quality check almost perfectly. A pilot study on gemma-2-2b shows that this check can hide a blind spot: on an object-tracking task the SAE's features bring back only 74% of the model's answer, and half of the missing quarter is a single hidden signal that only directions learned from that task recover. We propose to map such task-specific blind spots across more tasks and in Hindi and Hinglish, and to test whether the hidden signal is how the model links each object to its box.

1 What: the question and our claim

A language model turns text into **tokens** (words or word pieces) and processes them through a stack of **layers**; at each layer it holds an internal signal x of 2304 numbers per token. A **sparse autoencoder** (SAE) rewrites that signal as $x = \hat{x} + \varepsilon$: $\hat{x}$ is built from a few human-readable **features**, and ε, the **leftovers** (the SAE's error, sometimes called "dark matter" [4]), is what it cannot explain. A **direction** is one fixed pattern across those 2304 numbers. All terms are collected in Appendix D.

Research question. *When an SAE's features miss information a model is tracking, is the missing piece specific to the task, and is more of it missed in Indian languages?*

Our claim, from the pilot. SAE misses come in two kinds. Some are *concentrated and task-specific*: one or two directions in the leftovers, learned only from that task, bring back about half of what is missed (object tracking at layer 12; the running sum at layer 18). Others are *spread out* over dozens of directions (the names task at late layers). The standard average check sees neither. We propose to find out how common the first kind is and what it is:

Q1. Across tasks. Which tasks have a concentrated, task-specific gap, and is it tied to one prompt template or to a kind of computation?

Q2. Mechanism. In linking tasks, is the hidden signal the model's *binding* signal, which ties an object to its box [5]?

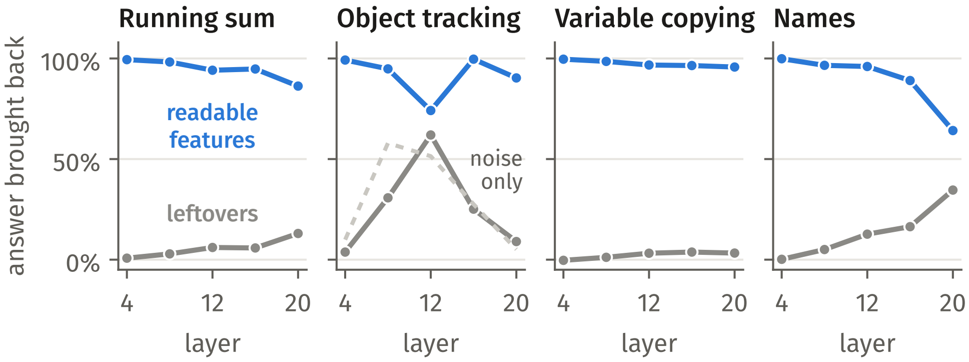
Q3. Across languages. Do SAEs miss more in Hindi and Hinglish than in English, and is it the same hidden signal?

2 Why it matters

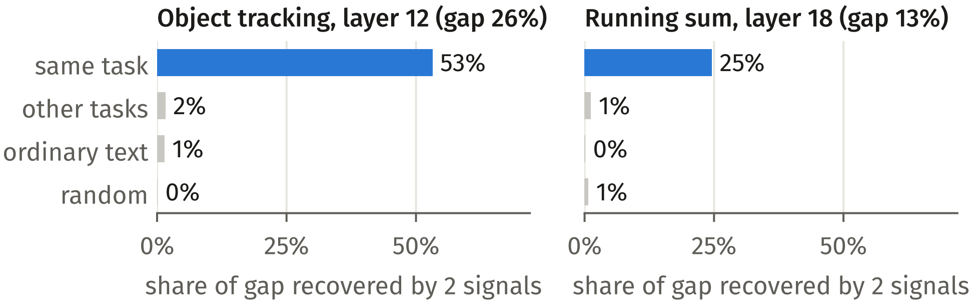
SAEs are used to explain, audit and steer models, and public suites such as Gemma Scope [8] are judged by a standard check: replace the model's signal with the SAE's version and see how much worse next-word prediction gets. On ordinary text the 16k Gemma Scope SAEs at layers 10 to 18 keep 98% to 99% of the model's performance. That number averages over ordinary text and says nothing about any one computation. Meanwhile the leftovers are known to matter: they are large and partly structured [4] and carry weight inside circuits [9]. If what SAEs miss were generic, one general fix (a better SAE, or a correction learned on ordinary text) would cover every task. If it is task-specific, an explanation of task T built only from features can be incomplete in exactly the way that matters, so every such explanation needs a per-task check; our deliverable is that check. Language raises the same question: the Gemma Scope SAEs were trained on predominantly English text [8]. Recent work finds language-specific and shared SAE features [1, 2], but does not measure what the leftovers carry in each language. Hindi speakers are among the users for whom SAE-based safety claims are least tested.

3 What we have done: the pilot study

Setup (Appendix B). Model: gemma-2-2b, Google's open 2-billion-parameter model [6]. SAEs: Gemma Scope, from 16k (16,384) to 262k features. Four tasks: a running sum ("Start with 2. Add 3. Add 1. Add 2. Total =" → 8), object tracking ("The key is in box B … The key moves to box



(a) Copying the features brings back most of the answer; the leftovers alone bring back little (dashed: the leftovers replaced by noise, for object tracking).



(b) Where the gap is concentrated, two directions learned from the same task close a real share of it; directions from other tasks or ordinary text close almost none.

Figure 1: The two pilot results behind the claim (16k SAEs, 150 test pairs). Details in Appendix B.4.

C … The key is in box” → C), variable copying (“t = 7. q = 4. x = q. q = 2. x =” → 4), and a well-studied names task as a control [12]. The three tasks we first planned failed: the model could not solve them (Appendix A). **The swap test** (a form of activation patching [13]): take prompts A and B that differ in one step, so their answers differ. Run the model on B, but at one layer copy in part of A’s signal, either its features $\hat{x}$ or its leftovers ε, and measure how far the answer moves from B’s to A’s (0% = not at all, 100% = fully, as when the whole signal is copied). Directions are learned on 300 training pairs and every score is measured on 150 different test pairs.

1. **Features carry most of it.** On the three tracking tasks, copying $\hat{x}$ brings back 74% to 100% at every layer tested (4 to 20); the leftovers alone 0% to 13%, except object tracking at layers 8 to 16 (25% to 62%), which is mostly an artefact: replacing the leftovers with noise already gives 28% to 58% (Figure 1a). The names control loses more late on (24% to 36% at layers 18 to 20).
2. **A concentrated gap.** For object tracking at layer 12 the features bring back 74% (95% CI 73% to 76%). Adding *one* direction from the leftovers, the most common pattern in them on training pairs, lifts this to 88%, about half of the gap; a random direction adds nothing, and even 32 directions learned on ordinary text add at most 1 point.
3. **Task-specific.** Two directions learned from the same task close 39% to 53% of that gap (two fitting methods); from other tasks −1% to 2%; from ordinary text 1% (Figure 1b). The running sum at layer 18 (gap 13%) behaves the same (25% to 58% vs at most 1%). The names loss is the other kind: about 32 directions are needed to close half of it at layer 18.
4. **Width shrinks it but does not close it.** From 16k to 262k features the object-tracking gap goes 26% → 20% → 24% → 23% → 15%, and one direction still closes 51% to 69% of what is left. Wider SAEs here also use more active features per token (78 vs 124), and the drop narrowly misses the bar we set before the run for a persistent gap (Appendix B.4), so width is an open question for Q1.
5. **Three known pitfalls** would each have given a false result: copying only the last token, probes that the prompt’s words alone can fool [7], and treating “readable” as “used” (Appendix B.5). The swap test we deliver guards against all three.

Limits. One model, one SAE family, synthetic tasks; width tested at one layer only.

4 How: plan and success criteria

Q1, across tasks. New tasks grouped by the computation they need: *linking* (objects in boxes; people and cities, e.g. "Asha lives in Pune. Ravi lives in Delhi. Asha lives in"), *counting* (longer running sums), *following references* (variable copying), plus the names control. For each task, layer and SAE width we measure the gap (100% minus what $\hat{x}$ brings back), how many directions close half of it, and a **specificity score** (share closed by same-task directions minus the best share from any other source). Three decisive tests: transfer between the two linking tasks (template or computation?), the ordinary-text control, and SAEs compared at matched numbers of active features (width or sparsity?).

Q2, binding test. Binding signals were found in other model families (LLaMA, Pythia) [5], so we first re-derive them on gemma-2-2b, compare them with the hidden direction, then edit the signal only along that direction and check whether the model names the other object's box.

Q3, across languages. The same tasks in Hindi (Devanagari), Hinglish (Latin script, code-mixed), Tamil or Bengali if the model can solve them, and French as a non-Indian control; since Hinglish also mixes in English words, we add the same Hindi sentences in Latin script without English words. Example, *illustrative and to be checked by a native speaker*: शुरुआत 2 से करें। 3 जोड़ें। 1 जोड़ें। 2 जोड़ें। कुल = → 8. Ordinary text for the standard check comes from IndicCorp v2 [3] and FineWeb-2 [10]. Design in Appendix C.

Table 1: Success criteria, fixed before running (95% confidence intervals throughout).

	Supported if	We change our mind if
Q1	gap ≥ 10%, and same-task directions close ≥ 30 points more of it than any other source	on ≥ 2 new tasks with a gap, other sources close as much as same-task directions
Q1 transfer	boxes→cities directions close ≥ half as much as the cities' own (a computation)	transfer near zero (a template)
Q2	editing along the direction sends ≥ 50% of pairs to the other box; a random direction ≤ 5%	the edit does no better than a random direction
Q3	Hindi: standard check ≥ 3 points lower, larger gap; English direction closes ≥ half as much as Hindi's own	Hindi's gap is no larger than English's

Deliverables. (1) A **blind-spot map**: gap, its kind and its specificity for every task, layer, width and language tested. (2) The **swap test** as a released script with a rule of thumb: if a task's gap exceeds 10%, keep the leftovers in its explanation. (3) A verdict on the binding hypothesis. **Resources and risks.** One 8 GB laptop GPU with the tested pilot pipeline (the three main pilot runs took about 6.3 hours). If binding signals cannot be re-derived on gemma-2-2b, Q2 keeps only the editing test; if the model fails Tamil or Bengali, we report that and keep Hindi and Hinglish; answers must be one token, so we fall back to letters or digits where names split.

5 When: timeline

When	Step	Output
Oct 19–26 (done)	Pilot study, four stages (Appendix B.2)	Concentrated gap found
Week 1 (Oct 27–Nov 2)	Q2: binding signals on gemma-2-2b, editing test; Q1 starts: new tasks, sparsity-matched widths	Binding verdict
Week 2 (Nov 3–9)	Q1: gap and specificity per task; Q3: Hindi and Hinglish tasks, can the model solve them?	Task map; usable languages
Weeks 3–4 (Nov 10–23)	Q3: standard check, gap, transfer per language; Tamil or Bengali if usable	Language map
Week 5 (Nov 24–30)	Buffer; more test pairs; robustness checks	Final figures
Week 6 (Dec 1–7)	Report and final presentation	Submission

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Appendices

The original plan and every detail of the pilot study

Roadmap. A: the original plan and why its tasks were dropped. B: the pilot study: setup, stages, measures, all results, pitfalls, verdict on the original hypotheses, caveats. C: multilingual design. D: glossary.

A The original position

The project began as *PRISM* (Probing Residual Information and State Manifolds), a test of a strong version of the "dark matter" worry: that SAEs are blind to the step-by-step state a model computes, so that this state lives in the leftovers. It made three hypotheses, each tied to one experiment (Table 2).

Table 2: The original hypotheses and how each was to be tested.

	Hypothesis	Experiment	Prediction if true
H1	The leftovers carry the task state	*Causal split:* copy $\hat{x}$ or ε from prompt A into a run on prompt B	copying ε brings back more of the answer than copying $\hat{x}$
H2	The state forms a simple, low-dimensional shape inside the leftovers	*Geometry:* the spread of ε across directions, probes for the state, low-dimensional projections	a probe reads the state from ε better than from $\hat{x}$ and better than from the prompt's words
H3	A few directions in the leftovers repair the SAE	*Repair:* $\hat{x}$ plus A's leftovers along their top r directions	a small r closes most of the gap; random directions do not

Original tasks. Three algorithmic tasks whose answer depends on a hidden running state: a sum with a remainder (modular arithmetic), the depth of nested brackets, and the size of the graph reachable from a node, plus the names task as a control. All share one format: prompts A and B that differ in one step, with a one-token answer.

Why they were dropped. The first overnight run (October 19) kept no usable pairs. A capability study (October 20; 128 pairs per setting, 0 to 8 worked examples, several difficulty levels) showed that gemma-2-2b does no better than simple shortcuts on all three: at the planned difficulty, accuracy was 13% vs a 21% shortcut (remainder sums), 41% vs 41% (bracket depth) and 30% vs 48% (graph size). Easier versions did not help: remainder sums and bracket depth stayed flat, and graph size improved but never became usable. The names control reached 100% with two worked examples, so the pipeline was sound. Three replacement tasks passed, alongside the names control (Figure 2).

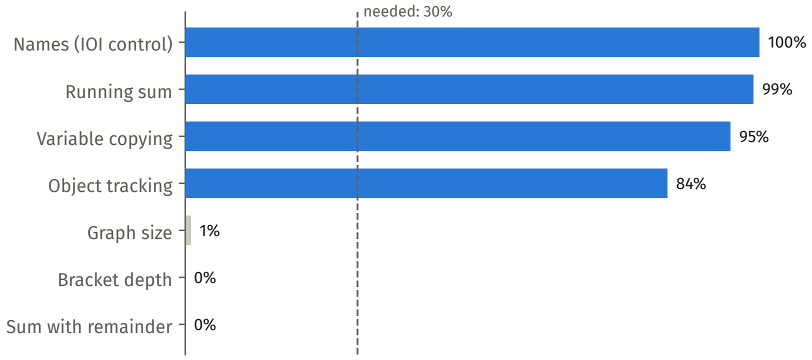

Figure 2: gemma-2-2b can do the four tasks we use, not the three first planned. Share of usable pairs in the capability study (model correct, and changing one step changes its answer; 30% needed). Used tasks at the difficulty we run, dropped tasks at their best setting.

B The pilot study

B.1 Setup

Table 3: Pilot setup.

Model	gemma-2-2b: 26 layers, internal signal of 2304 numbers per token, 16-bit (bf16) precision
SAEs	Gemma Scope residual-stream SAEs with a JumpReLU activation [8, 11] (`gemma-scope-2b-pt-res-canonical`), widths 16k, 32k, 65k, 131k, 262k
Standard check	16k SAEs at layers 10, 12, 14, 18 keep 99.1%, 98.7%, 98.5%, 98.0% of next-word performance on ordinary text (the Pile), measured against replacing the signal with its average
Pairs	300 training and 150 test pairs per task and setting, no prompt shared; all fitting on training pairs only
Pair filter	running sum, variable copying, names: the model must give both answers correctly and with a clear margin; object tracking (harder for the model): A's answer must be correct and the one-step change must shift the model's preference by a clear margin
Statistics	95% paired bootstrap confidence intervals
Code checks	78 unit tests; matches Hugging Face's implementation of the model (same top answer on 95% of prompts)
Compute	one 8 GB laptop GPU (RTX 5060)

B.2 Stages

Table 4: The four stages of the pilot.

Date	Stage	What ran	Outcome
Oct 19	0. Overnight run	original three tasks, all three experiments	no usable pairs; no results
Oct 20	1. Checks	code checked against Hugging Face; capability study of 7 tasks	original tasks dropped; three replacements chosen
Oct 25	2. Quick pilot	24 min; 160 training / 80 test pairs; layers 6, 12, 18; 16k SAEs	H1 rejected; a small, concentrated gap in object tracking and the running sum
Oct 26	3. Main pilot	three runs, about 6.3 h: swap test at layers 4 to 20; repair and specificity at layers 12 to 18; widths 16k to 262k at layer 12	results below

The quick pilot and the main runs agree: object tracking at layer 12 gave 75% vs 74% (features) and 61% vs 62% (leftovers), and one direction closed 51% vs 52% of the gap; the running sum's gap at layer 18 was 12% vs 13%.

B.3 Measures

Share of the answer brought back (NLDR, normalized logit-difference recovery). A *logit* is the model's raw score for a possible next token. With D = logit of A's answer minus logit of B's answer, NLDR $= (D_{\text{copied}} - D_B)/(D_A - D_B)$, averaged over test pairs: 0 = nothing brought back, 1 = fully. **Copy conditions:** the whole signal (control, must give 1); $\hat{x}$ only (A's $\hat{x}$, B's ε); ε only (A's ε, B's $\hat{x}$); noise (B's $\hat{x}$ plus random noise of the same size as ε, which erases B's leftovers). **Positions:** copying every token from the first one that differs (used everywhere except pitfall 1) or the last token only. **Repair:** $\hat{x}$ plus A's leftovers projected on r directions U_r, i.e. $\hat{x} + U_r U_r^\top \varepsilon$. Directions come from the task's own training leftovers (*unsupervised*: the most common patterns; *supervised*: the patterns that best separate the answers), from ordinary text, from another task (*transfer*), or at random. **Gap** $= 1 - \text{NLDR}(\hat{x}\text{ only})$; **share of the gap closed** = (repair − $\hat{x}$ only)/gap. **Probes:** simple linear classifiers trained to read the answer from the signal at the last token, compared with a probe that sees only the prompt's words.

B.4 Results

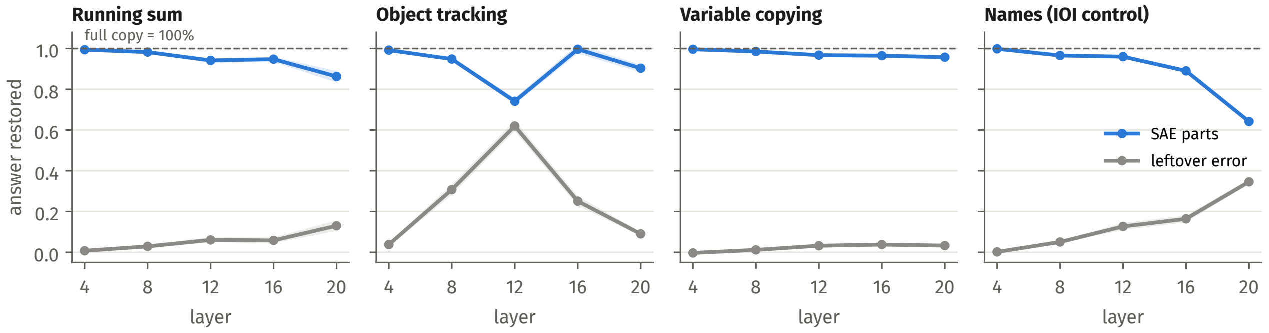

Figure 3: Features (“SAE parts”, blue) bring back most of the answer at every layer; the leftovers (“error”, grey) little, except object tracking in the middle layers (copy from the first changed token; 95% intervals).

Table 5: Share of the answer brought back (%) by copying the features / the leftovers, per layer.

	layer 4	layer 8	layer 12	layer 16	layer 20
Running sum	99 / 1	98 / 3	94 / 6	95 / 6	86 / 13
Object tracking	99 / 4	95 / 31	74 / 62	100 / 25	90 / 9
Variable copying	100 / 0	99 / 1	97 / 3	96 / 4	96 / 3
Names (control)	100 / 0	97 / 5	96 / 13	89 / 16	64 / 35

Object tracking’s leftovers score is inflated: replacing the leftovers with noise alone gives 28% to 58% at layers 8 to 16 (51% at layer 12), because it erases B’s leftovers; A’s leftovers add about 11 points at layer 12.

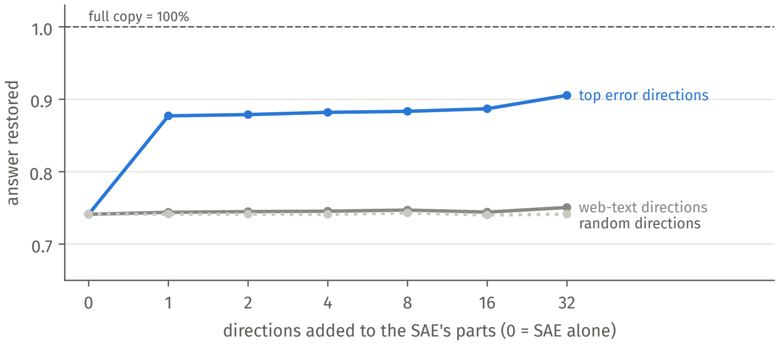

Figure 4: Object tracking at layer 12: one direction from the leftovers lifts 74% to 88%; directions learned on ordinary text (“web text”) or at random add nothing, even 32 of them.

Table 6: Object tracking, layer 12: share brought back (%) by $\hat{x}$ plus N directions.

N	0	1	2	4	8	16	32
top directions of the leftovers	74	88	88	88	88	89	91
ordinary-text directions	74	74	74	75	75	74	75
random directions	74	74	74	74	74	74	74

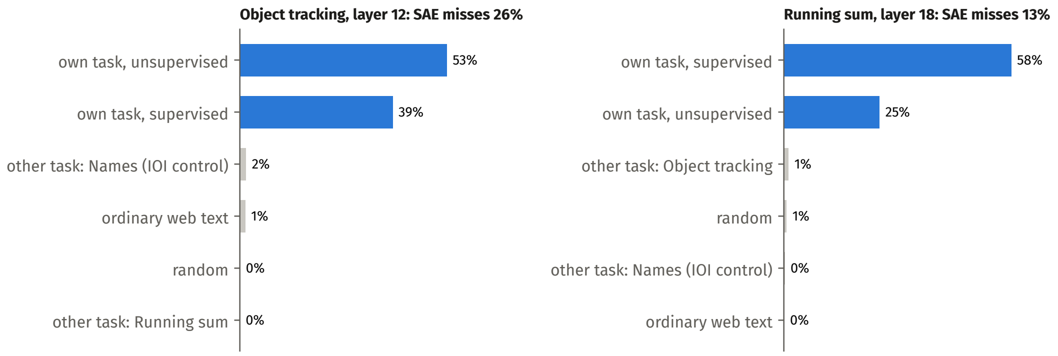

Figure 5: Only directions learned from the same task close a real share of the gap (2 directions per source, 16k SAE; "ordinary web text" is the Pile).

Table 7: Share of the gap closed (%) by 2 directions from each source.

Task (gap)	Source	Closed
Object tracking, layer 12 (26%)	same task, unsupervised / supervised	53 / 39
	other task: names / running sum	2 / −1
	ordinary text / random	1 / 0
Running sum, layer 18 (13%)	same task, supervised / unsupervised	58 / 25
	other task: object tracking / names	1 / 0
	ordinary text / random	0 / 1

The names control is the other kind of gap: it loses 24% at layer 18 (36% at layer 20), and about 32 same-task directions are needed to close half of it at layer 18.

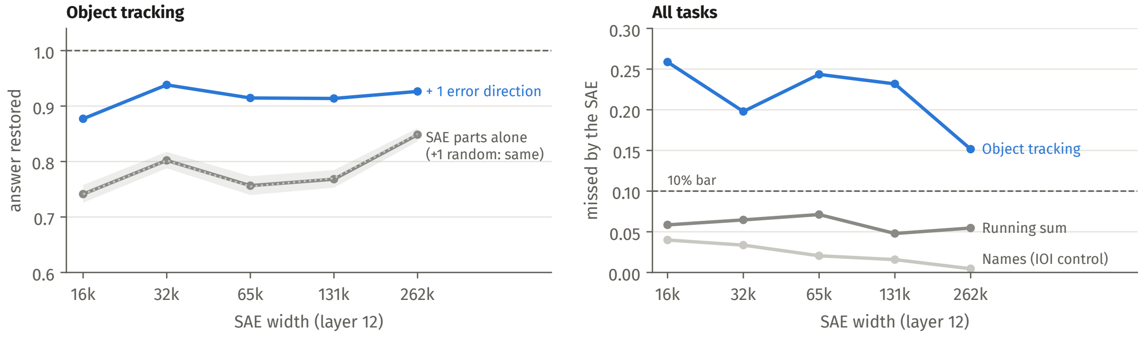

Figure 6: Wider SAEs at layer 12. Left: object tracking with the features alone (grey) and with one direction from the leftovers (blue). Right: the gap for three tasks.

Table 8: Width sweep at layer 12 (%). "Active" = features active per token.

Task	Width	$\hat{x}$ only [95% CI]	gap	+1 leftover dir.	+1 random dir.	active
Object tracking	16k	74 [73, 76]	26	88	74	78
	32k	80 [79, 82]	20	94	80	75
	65k	76 [74, 77]	24	91	76	76
	131k	77 [75, 78]	23	91	77	131
	262k	85 [84, 86]	15	93	85	124
Running sum	16k / 262k	94 / 95	6 / 5	94 / 94	94 / 95	80 / 129
Names (control)	16k / 262k	96 / 100	4 / 0	96 / 100	96 / 100	89 / 139

Across all widths the running sum's gap stays between 5% and 7% and the names gap at or below 4%. Before the run we set a rule: a gap counts as persistent if the widest SAE keeps at least 60% of the 16k gap. Object tracking keeps 15.2% of 25.9%, just under the 15.5% bar, so by our own rule the width result is borderline; we test it again with sparsity matched (Q1).

B.5 Three pitfalls

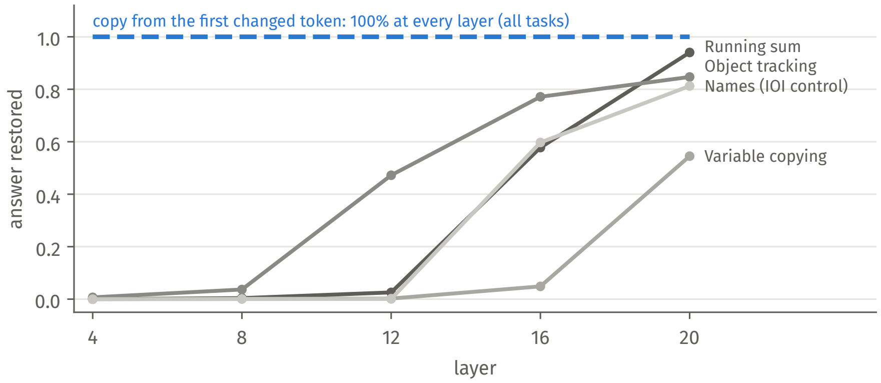

Figure 7: Pitfall 1: copying the whole signal at the last token only brings back little in early and middle layers; copying from the first changed token brings back 100% at every layer.

Pitfall 1, copying only the last token. In early and middle layers the state still sits on earlier tokens, so copying only the last token brings back little even with the *whole* signal (%, layers 4/8/12/16/20): running sum 0/0/3/58/94; object tracking 1/4/47/77/85; variable copying 0/0/0/5/55; names 0/0/0/60/81. Copying from the first changed token brings back 100% at every task and layer [cf. 13].

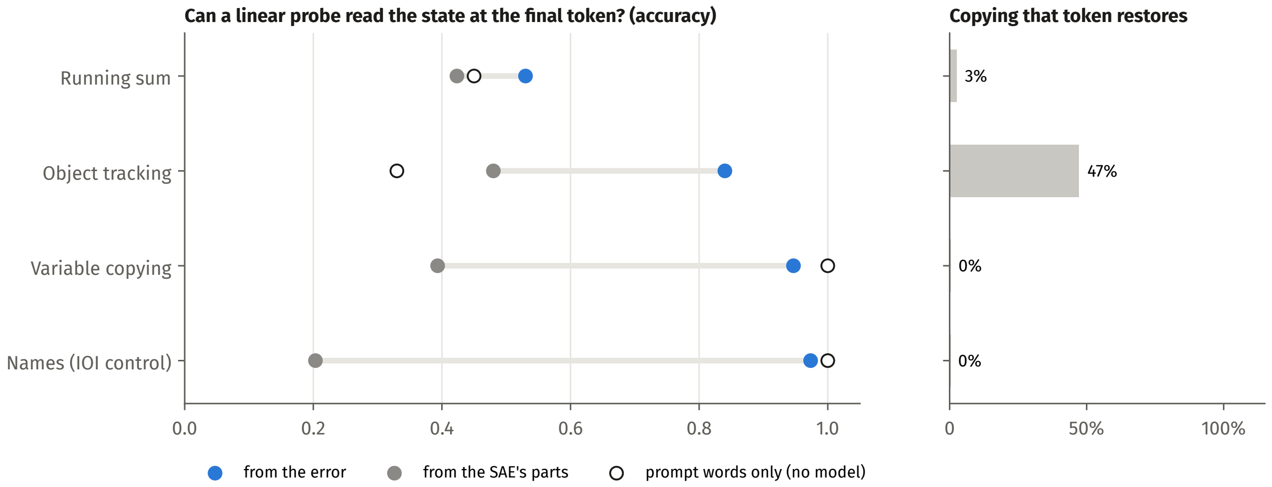

Figure 8: Pitfalls 2 and 3 at layer 12: probes read the state from the leftovers better than from the features, but the prompt's words alone often do as well, and copying that token brings back little.

Table 9: Probe accuracy at the last token, layer 12 (%), and what copying that token brings back.

	probe on ε	probe on $\hat{x}$	words only	chance	last-token copy
Running sum	53	42	45	25	3
Object tracking	84	48	33	18	47
Variable copying	95	39	100	11	0
Names (control)	97	20	100	11	0

Pitfall 2, the prompt gives it away: a probe that sees only the prompt's words scores 100% on variable copying and names, so a high probe score there proves nothing [7]. **Pitfall 3, readable is not used:** for object tracking and names, probes read the state far better from ε than from $\hat{x}$ (84%

vs 48%; 97% vs 20%), yet copying that token brings back at most 47%: the model does not use it there yet.

B.6 Verdict on the original hypotheses

Table 10: What the pilot says about each original hypothesis.

H1 (leftovers carry the state)	**Rejected** for every task: $\hat{x}$ beats ε at every layer.
H2 (simple shape in the leftovers)	**Not supported as stated**: the state can be read from ε but is not used there, and for two tasks the prompt's words alone give it away.
H3 (a few directions repair)	**Supported, narrowly**: one direction closes about half of object tracking's gap, two close 58% of the running sum's; the directions are task-specific. This became the project's claim.

B.7 Caveats

- One model (gemma-2-2b), one SAE family (Gemma Scope), synthetic tasks; width tested at layer 12 only.
- Object tracking is the hardest task for the model: in the main runs only 27% of generated candidate pairs passed its pair filter (40% to 100% for the others), so its pairs are the most selected. Figure 2 uses the capability study's sample and criteria, hence its higher 84%.
- The names control's late-layer loss (24% to 36% at layers 18 to 20) is spread over many directions: part of the late-layer loss is a general SAE weakness.
- The 131k and 262k SAEs also use more active features per token (124 to 139 vs 78 to 89 at 16k), so part of the shrinking gap may come from that rather than from width.
- The binding explanation is a hypothesis, and the multilingual work is a plan with no results yet.

C Multilingual design

Table 11: Multilingual hypotheses (criteria as in Table 1).

ID	Hypothesis	Measure	Supported if
M1	SAEs explain Hindi text worse	standard check on ordinary text, per language	at least 3 points below English (98% to 99%)
M2	SAEs miss more task state in Hindi	gap and specificity per language	larger gap than English, non-overlapping intervals
M3	The hidden signal is shared across languages	direction learned in English, used on Hindi	closes at least half as much as Hindi's own

Languages: English (baseline); Hindi (Devanagari; the best-resourced Indian language, most likely to work); Hindi in Latin script without English words, and Hinglish (Latin script, code-mixed), which together separate script from vocabulary; Tamil or Bengali if the model can solve the tasks; French as a high-resource non-Indian control.

Templates. The answer must be the last token, so verb-final Hindi uses a question plus "answer" format. All examples are *illustrative, to be checked by a native speaker and against the tokenizer*:

- Running sum (Hindi): शुरुआत 2 से करें। 3 जोड़ें। 1 जोड़ें। 2 जोड़ें। कुल = → 8
- Object tracking (Hindi): सिक्का डिब्बे B में है। चाबी डिब्बे E में है। चाबी डिब्बे C में चली जाती है। चाबी किस डिब्बे में है? डिब्बा → C
- Names (Hinglish): `Jab Rahul aur Priya market gaye, Priya ne ek book di. Book kisko mili? Answer:` → Rahul
- Variable copying is code-like and identical in every language: a language-neutral control.

Constraints: prompts A and B need the same number of tokens and a one-token answer; many Hindi names split into several tokens, so names are picked from single-token ones (the pipeline already drops and counts pairs that break this). **Steps:** (1) capability check per language (at least 30% usable pairs; failing is itself a finding); (2) the standard check on IndicCorp v2 and FineWeb-2 text (M1); (3) swap test, gap, repair and specificity (M2); (4) the English direction used on Hindi (M3).

D Glossary

Token	a word or word piece; the unit the model reads
Layer	one of gemma-2-2b's 26 processing stages
Internal signal (residual stream)	the model's running representation at one layer: 2304 numbers per token
Sparse autoencoder (SAE)	a tool that rewrites the internal signal as a few active, human-readable features
Features, $\hat{x}$	the SAE's reconstruction from its active features ("SAE parts" in the appendix figures)
Leftovers, ε	$x - \hat{x}$: what the SAE cannot explain (its "error", or "dark matter")
Width	how many features an SAE can choose from (16k = 16,384 up to 262k)
Active features (L0)	how many features are switched on for one token
Standard check	how much next-word performance survives when the SAE's version replaces the signal
Prompts A and B	two prompts that differ in one step, so the right answer differs ("clean" and "corrupt")
Swap test (activation patching)	run on B but copy part of A's signal at one layer
Logit, NLDR	a logit is the model's raw score for a next token; NLDR is the share of the answer brought back (0 to 1)
Gap	the share the features miss: one minus NLDR of copying $\hat{x}$ only
Direction	one fixed pattern in the 2304 numbers; "N directions" = an N-dimensional slice of the signal
Unsupervised / supervised directions	the most common patterns in the leftovers / the patterns that best separate the answers
Specificity score	share closed by same-task directions minus the best share from any other source
Probe	a simple classifier trained to read a value (e.g. the box) from the signal
Binding	how a model links an entity to its attribute (an object to its box)
IOI	"indirect object identification", the names task [12]
Hinglish	Hindi written in Latin script and mixed with English words
95% CI	bootstrap confidence interval: the range the result would likely fall in on new test pairs